

Name: Ujwala Pavuluri
Email: nagap2025@gmail.com
LinkedIn: <https://www.linkedin.com/in/ujwalapavuluri/>
Mobile: +1832-599-7009



Lead Full Stack Developer

SUMMARY

- With more than 14 years of practical experience in software engineering, including enterprise and application architecture, design, and development, I specialize in cloud-native development, distributed applications, microservices, event-driven architecture, and legacy migration.
- Expert in developing multi-tier enterprise-level web applications using various Java/J2EE technologies including JSP, Spring, EJB, JDBC, Java Beans, and Web Services.
- Extensive experience in working with Frameworks - Spring, JSF, Hibernate, and MVC.
- Expert in Spring Framework including configuring Spring Application Context with Dependency Injection, Spring MVC, Spring Boot, and Spring Batch.
- experience in designing User Interface (UI) applications and professional web applications using HTML5, CSS3, JavaScript, Bootstrap, React.js, AngularJS, Ext.js, ECMA Script, TypeScript, DOM, jQuery, Ajax, XML, JSON, and Node.js.
- Worked consuming APIs using Axios and manipulating/ consuming JSON objects.
- Extensive experience working with Redux-saga library to manage side-effects within the system, perform asynchronous calls using generator functions in ES6 and access browser cache.
- Extensive experience working with JavaScript Frameworks like React JS, Node JS, Redux.
- Skilled in creating Web Applications, User Interfaces and Layouts with HTML, Typescript, CSS3 and JavaScript.
- Contributed to all levels of the user experience and in all phases of the product development lifecycle, from defining a product's overall architecture and structure.
- Expertise in designing, developing and deploying applications using J2EE tech including spring, Hibernate, Struts, AJAX, and Webservices.
- Proficient in developing front-end systems with JavaScript, JQuery, Ajax, Bootstrap, HTML5, CSS3 and JavaScript frameworks such as AngularJS.
- Worked with React.JS components, Forms, Events, Keys, Router, Animations and Flux concept.
- Expertise in Implementing Java/J2EE design patterns like MVC, Session Facade, Data Access Object (DAO), Factory, Singleton, and Front Controller.
- Strong experience in XML-related technologies including XSD, DTD, XSLT, XPATH, DOM, SAX, JAXP, JAXB, XML-RPC, and XML Beans.
- Worked on Front end by using AngularJs, React JS & JavaScript.
- Experience in using IDE Tools like Eclipse, NetBeans, and Rational Application Developer (RAD) for application development.
- Experience in using HTML5/DHTML, XHTML, JavaScript, JQuery, JSON, XML, XSL, XSLT, CSS3, and Applets/Swings.
- Experience with JavaScript Frameworks like Angular 6/4/2, Ext.JS, and Node. JS.
- Experience in using CSS with Angular.JS which is called in different scenarios required such as moving to the next page or animations while page and screen changes.
- Experience in designing and implementing Microservices architecture, implementing RESTful APIs, and working with GraphQL for efficient data retrieval.
- Experience with Java Multi-threaded applications, Drools, JBPM, and Oracle ATG Commerce
- Strong knowledge and good experience in providing logging support using Log4j.

- Experience in Amazon Web Services like EC2, S3 bucket, ELB, Auto-Scaling, SNS, SQS, AMI, IAM, Dynamo DB, Elastic Search, Cloud Formation, Virtual Private Cloud (VPC) through AWS Console, and API Integration.
- Proficient in handling and administering Web servers/application servers IBM Web Sphere, Apache Tomcat and JBoss.
- Extensively used building tools like ANT, and Maven to compile, package and deploy the components to the Application Servers.
- Experience in shell, ruby and Python scripting. Automated processes with custom built Python & Shell scripts.
- Expertise in producing and consuming SOAP as well as REST web services using WSDL, SOAP, JAXWS, and JAXRS.
- Experience with Agile (Scrum) and test-driven development and version control (SVN, Git, etc.). Keen focus on consistency and attention to detail primarily related to coding standards.
- Strong experience in database design using PL/SQL to write tables, Stored Procedures, Functions, Triggers, and Indexers and proficiency in writing complex queries, using Oracle, SQL Server, MySQL, PostgreSQL, and MongoDB/NoSQL.
- Strong experience in client interaction and understanding business applications, business data flow, and data relations from them.
- Experience with RESTful Microservices development with Spring Boot, AWS API Gateway
- Good knowledge on Caching, Clean code, NoSQL databases including MongoDB for the cluster management with Apache KAFKA as messaging system.
- Committed to excellence, self-motivator, fast learner, team player, ability to deal with people diplomatically, and a prudent developer with strong problem-solving skills and communication skills.
- Experienced in Oracle, RDBMS, SQL Server, PostgreSQL12, MySQL, MongoDB, Sybase, and DB2 as the Database.
- Experienced in JBoss, Tomcat, Web Sphere, Web Logic, and ATG Dynamo application servers.
- Experienced in all phases of Software Development Life Cycle (SDLC), RUP, Agile i.e., planning, architecture, Use Case analysis, Object Oriented design, development, debugging, Unit Test, Integration, managing and monitoring of complex, scalable, secured, optimized, large data-intensive enterprise systems.
- Strong GUI experience with JHTML, JSP, JavaScript, HTML, DHTML, jQuery, AJAX, and CSS.
- Experienced in Linux, UNIX & Windows NT/XP/2000 Professional platforms.
- Experienced leading small to mid-sized development teams.

TECHNICAL SKILLS

Languages	Java, PL/SQL, Python, GraphQL, C++, C
Java/J2EE Technologies	Java, JSP, Servlets JDBC, JNDI, JMS, JSTL, Java Beans, RMI, Java Multithreading, Generics, and Collections, EJB, Tiles
Web Technologies	HTML5, XML, DOM, CSS3, JavaScript, XPath, AJAX, jQuery, Angular.js, Angular 6/4/2, Bootstrap, JHipster, React
Methodologies	UML, Agile, Waterfall
Message Queue	Apache Kafka, Open Refine, Hazelcast, Redis
CI/CD	Git, BitBucket, SVN, Jenkins, JFrog, Docker, Kubernetes, AWS, Terraform
Static Code Analysis	SonarQube, Find Bugs, Simian, Source Monitor, PC-Lint
Databases/Tools	Postgres, MySQL, GraphQL
Security Tools	Veracode, Checkmarx
Wireframes	Balsamiq, Figma
Chaos Engineering	Chaos Mesh
Performance Engineering	Gatling, Jmeter
Observability	OpenTelemetry, Appdynamics, Datadog
Cloud	AWS SQS, SNS, S3, Lambda, EC2, ECK
Open-Source Scanning	Palamida, BlackDuck

EDUCATION

- JawaharLal Nehru Technological University - Hyderabad, Telangana, India 2009
- International Institute of Information Technology - Hyderabad, Telangana, India 2011

EXPERIENCE

Verizon – New York,US
Lead full stack developer

Apr 2024 to Till Date

Project:

Serving as a Solution Architect within the Enterprise Architecture and Center of Excellence team, I have driven innovation in mobile marketing and led the architecture and design of platforms that deliver seamless omni-channel experiences for Verizon customers. My work spans across retail stores, indirect channels, mobile apps, and web applications, ensuring cohesive and customer-centric solutions.

Key Responsibilities and Achievements:

- Conducted an in-depth analysis of current mobile marketing strategies, identifying opportunities to improve mobile app engagement and optimize marketing efforts through innovative solutions.
- Developed the Madme platform to enhance push notification effectiveness, which strategically identifies optimal moments for sending notifications, significantly improving click-through rates.
- Implemented the Branch platform to facilitate seamless cross-channel marketing, overcoming limitations imposed by platforms like Meta and Instagram and enabling broader reach and more efficient marketing across multiple channels.
- The use of Branch led to improved user acquisition strategies and higher retention rates, contributing to the overall success of marketing campaigns.
- Worked extensively with AngularJS, developing dynamic views, managing two-way data binding, and rendering components using Angular directives and custom HTML tags.
- Designed and implemented various AngularJS components, including forms, event handlers, routers, animations, and state management, ensuring a modular and maintainable front-end architecture.
- Focused on building a components library with reusable elements such as Tree, Slide-View, and Table Grid to be used across multiple pages and applications.
- Extensively worked on building n-tier applications using Java, J2EE, MVC Framework, Spring Boot, Spring, Hibernate, JavaScript, HTML/HTML5, CSS3, AngularJS, Oracle, and SQL.
- Implemented AngularJS services and factories for data management, ensuring seamless interaction between the UI and back-end services.
- Leveraged AngularJS for templating, enabling faster compilation and efficient rendering of dynamic content with directives and controllers.
- Worked on Webpack to manage build processes, executing linters, pre/post-processors, tests, and transpilers based on build targets, improving overall performance.
- Led distributed test automation using Jenkins as part of a Continuous Integration process, integrating Sonar for code quality checks and ensuring smooth deployment pipelines.
- Created end-to-end integration tests with Cypress and unit tests using Jest, ensuring a reliable and stable system across all environments.
- Enhanced user experience by integrating location-based services using AngularJS and Google Maps API for location search functionality.
- Validated solutions through rigorous Proof of Concept (POC) testing, exploring potential improvements and optimizations for system reliability.
- Designed a roadmap for transitioning from POC to a fully stable platform, using insights gained from testing to ensure scalability and long-term success.

- Addressed critical limitations within the existing system, focusing on performance improvements and system optimization through innovative approaches and architecture.

Key Technologies and Tools:

- Languages/Frameworks: AngularJS, Java, Spring Boot, Hibernate, J2EE, JavaScript, HTML5, CSS3
- Cloud: Google Cloud Platform (GCP)
- Mobile Marketing: Madme, Branch
- APIs: Google Maps API, GitHub API
- State Management: Angular Services, Directives, Factories
- CI/CD & DevOps: Jenkins, Sonar, Webpack
- Testing: Cypress, Jest, End-to-End Testing
- Front-End: AngularJS, NPM, SCSS, Webpack

John Deere – Moline, Illinois	Jan 2023 to Feb 2024
Lead full stack developer	

Project:

Overhauled Dealer Management and Ordering System (JohnDeere)

The transformation of a monolithic dealer management and ordering system to a cloud-native, scalable, microservices-based architecture using AWS technologies, ensuring improved agility, reduced time-to-market, and a competitive edge in the market.

Key Responsibilities and Achievements:

- **Cloud Migration and Architecture Modernization:**
Spearheaded the transition of the system from a traditional monolithic architecture to a cloud-native microservices architecture using AWS services. Leveraged Amazon Elastic Kubernetes Service (EKS) for container orchestration, Amazon SQS and Amazon SNS for messaging, and Amazon S3 for scalable and reliable data storage. Applied cloud best practices to ensure a robust, highly available, and scalable solution.
- **Serverless Design and Implementation:**
Applied a serverless architecture using AWS Lambda, AWS API Gateway, and Amazon DynamoDB to optimize resource usage and improve scalability. This included designing event-driven workflows and automating processes through AWS CloudFormation for infrastructure-as-code (IaC) deployments.
- **Microservices Strategy and Tools Selection:**
Analyzed existing systems and gaps in developer capabilities, providing a clear migration strategy from monolithic to microservices. Selected and integrated the necessary tools and technologies for a successful migration, aligning business and technical goals for a seamless transition. Key services included AWS S3 for storage, AWS EventBridge for event management, and AWS Glue for ETL (Extract, Transform, Load) processes to modernize data workflows.
- **Proof of Concept (PoC) Development:**
Designed and executed multiple PoCs to validate the feasibility of cloud migration, using technologies like AWS Lambda and Amazon Kinesis for real-time data processing. These PoCs received management approval and laid the foundation for the full-scale migration.
- **State Management and Scalable Front-End Architecture:**
Developed a modern, scalable front-end architecture with AngularJS, ngrx, and RxJS for reactive state management, ensuring dynamic, real-time user interfaces. Integrated AWS API Gateway for handling API requests and AWS Lambda for backend services, creating a seamless front-end and back-end communication layer.
- **CI/CD Automation with Jenkins and AWS:**
Established a fully automated CI/CD pipeline using Jenkins and AWS CodePipeline, enabling faster, more

reliable software releases. The pipeline ensured seamless deployment to AWS services like Elastic Beanstalk and Lambda, significantly reducing manual intervention in the deployment process.

- **Security and Compliance:**
Implemented best practices for security and compliance, focusing on access control (using IAM roles and policies), encryption (using AWS KMS), and networking (using Amazon VPC and private subnets) to maintain high standards for data protection and system security in the cloud.
- **Testing and Performance Optimization:**
Developed automated tests using JUnit, Gatling, and Swagger to ensure code quality and system performance. Leveraged AWS CloudWatch and AWS X-Ray for monitoring and debugging application performance, ensuring smooth operation and quick issue resolution.

Key Technologies and Tools:

- **Cloud Technologies:** AWS Lambda, API Gateway, DynamoDB, S3, CloudFormation, EventBridge, Glue, Kinesis, SQS, SNS, IAM, VPC, KMS
- **Languages/Frameworks:** Node.js, Python, Java, Spring Boot, AngularJS, JavaScript, ES6, React
- **State Management:** ngrx, Redux, RxJS
- **CI/CD & DevOps:** Jenkins, AWS CodePipeline, AWS Elastic Beanstalk, AWS CodeBuild, Kubernetes
- **Testing:** JUnit, Gatling, Swagger, Insomnia
- **Version Control:** GitHub
- **Database:** PostgreSQL, Amazon DynamoDB

Falabella –CL, Santiago
Lead full stack Developer

Jun 2021 to Dec 2022

Project:

Enhanced Data Architecture & State Management for Order Management System

Architected and designed a scalable, efficient state management solution to handle complex order management workflows across microservices, improving data accessibility, operational efficiency, and system performance.

Key Responsibilities and Achievements:

- **Data Mesh Architecture for Scalability:**
Implemented the Data Mesh architecture to improve data handling, scalability, and decentralization of data ownership across teams. Applied AWS technologies such as Amazon S3 for data storage, AWS Glue for ETL processes, and Amazon Redshift for analytics, ensuring seamless integration of decentralized data sources across the organization.
- **Custom State Management Library:**
Designed and developed a custom state management library to handle complex state transitions across multiple microservices in a distributed architecture. Integrated the library with AWS Lambda to process state changes and with Amazon SQS for communication between services, automating state transitions and ensuring smooth operations across the system.
- **Seamless Integration with External Systems:**
Facilitated integration with external systems (such as carriers and vendors) through AWS API Gateway, ensuring smooth communication between internal services and third-party systems. Leveraged Amazon SNS and Amazon SQS for reliable, asynchronous messaging between services, enhancing system reliability and performance.
- **Microservices Architecture and AWS Integration:**
Played a key role in designing the system architecture, transitioning the platform to a microservices-based environment using AWS Elastic Kubernetes Service (EKS) for container orchestration and Amazon RDS for

database management. This ensured improved scalability, resilience, and performance across all microservices.

- **Optimized State Management:**
Developed a custom solution to track and manage order states across all microservices using AWS DynamoDB for state persistence. This solution ensured faster, more reliable order state updates and minimized manual intervention, automating the process across all services.
- **Automation and CI/CD Integration:**
Integrated AWS CodePipeline and AWS CodeBuild for continuous integration and deployment of the state management library. This automated the deployment process, improving the speed and reliability of updates to the system.
- **UI Development and Front-End Architecture:**
Developed UI screens using React.js, Redux, and React-Bootstrap, with state management handled by Redux for dynamic data flow and communication between the front-end and back-end services. Integrated AWS API Gateway and AWS Lambda to handle API calls, ensuring seamless data flow between the user interface and microservices.
- **Enhanced Integration with Third-Party Services:**
Ensured that the custom state management solution could easily integrate with third-party services, such as logistics carriers and vendors, using AWS API Gateway and Amazon SQS to provide scalable and secure communication channels.
- **Improved Operational Efficiency:**
Automated state management across microservices, reducing the need for manual intervention and improving system consistency. The custom library allowed for better tracking, faster updates, and reduced error rates in order state transitions.
- **Scalability and Future-proofing:**
Designed the state management library with flexibility in mind, allowing for future enhancements and easy integration with additional system components. This approach ensured that the platform could scale as the system grew and new services were added.

Key Technologies and Tools:

- **Cloud Technologies:** AWS Lambda, API Gateway, DynamoDB, S3, SQS, SNS, Glue, Redshift, EKS, RDS, IAM
- **Languages/Frameworks:** React.js, Redux, JavaScript, ES6, Node.js, Python
- **State Management:** Redux, ngRx
- **CI/CD & DevOps:** AWS CodePipeline, AWS CodeBuild, EKS, CloudFormation
- **Testing & Debugging:** JUnit, Swagger, Postman, AWS CloudWatch, AWS X-Ray
- **Version Control:** GitHub
- **Database:** Amazon RDS, DynamoDB

Urgently –California,US
Lead full stack developer

Dec 2020 to Jun 2021

Project:

Designed and implemented microservices architecture, resulting in a 40% decrease in system downtime and improved overall system reliability for Partner management and Integrations.

Responsibilities:

- Architected a microservices-based solution, decoupling monolithic components to improve system scalability,

maintainability, and fault tolerance.

- Focused on enhancing Partner Management and Integrations systems to ensure high availability and efficient communication across multiple services.
- Reduced system downtime by 40%, increasing overall system reliability and uptime, directly contributing to improved business operations.
- Developed user interface by using the React JS, Flux for SPA development.
- Implemented React JS code to handle cross-browser compatibility issues in Mozilla, IE, Safari and FF.
- Used React-Router to turn the application into Single Page Application
- Worked in using React JS components, Forms, Events, Keys, Router, Animations and Flux concept.
- Involved in building APIs and Views utilizing Python to build an interactive web-based solution.
- Responsible for creating an instance on Amazon EC2 (AWS) and deployed the application on it.
- Indexed documents using Elastic Search.
- Developed Open stacks API to Integrate with Amazon EC2 cloud-based architecture of AWS, including creating machine Images
- Optimized critical components, ensuring smooth integration with external partners, reducing the risk of disruptions and enhancing system performance.
- Introduced circuit breakers, service discovery, and load balancing to ensure resiliency across microservices, minimizing service interruptions.
- Implemented automated monitoring and alerting systems to quickly identify and address potential issues, improving proactive response times.
- Implemented CI/CD pipelines using Jenkins and GitHub, ensuring streamlined deployment and faster delivery of updates to production without compromising stability.
- Employed JUnit and JMeter for testing and ensuring the quality of individual services and the overall system.

Environment: Java 8, Springboot, MySQL, Junit, Jmeter, Swagger, Postman, Github, Jenkins

Morgan Stanley –New York,US

Apr 2020 to Dec 2020

Lead full stack developer

Project:

Caching solution for Portfolio Accounting & Management - Distributed caching using Hazelcast for substituting information from legacy, very long-time-consuming stored procedures.

Responsibilities: -

- Architected a distributed caching solution using Hazelcast to cache critical data and reduce dependency on legacy, resource-intensive stored procedures.
- Improved system performance by drastically reducing data retrieval times, enhancing overall application responsiveness and scalability.
- Built stable React components and stand-alone functions to be added to any future pages.
- Used predefined components from NPM and redux library.
- Used React Router to turn the application into a single page application and used React Components, Forms, Events, Keys, and Router concept.
- Created and used Reducers and Actions, maintained states in stores, and dispatched the actions using redux.
- Replaced time-consuming stored procedures with a fast, efficient caching layer, reducing the load on legacy systems and improving the speed of data access for Portfolio Accounting & Management.
- Optimized the data flow and reduced processing times, increasing the system's efficiency and responsiveness, especially for high-frequency queries.
- Designed the caching solution for high scalability and fault tolerance, ensuring that the solution could handle large volumes of data and traffic without degrading system performance.
- Ensured smooth integration with existing services and databases, maintaining data consistency across distributed nodes.
- Implemented monitoring tools to track the performance of the caching layer, ensuring proper cache invalidation and data freshness.
- Established guidelines for cache management to optimize memory usage and minimize cache misses.

Environment: Java 8, Springboot, MySQL, Junit, Jmeter, Swagger, Postman, Github, Jenkins , Hazel cast

Project:

A master data management like tool meant to create a 360-degree view of data by creating a data lake and integrating various applications across the organization.

Match Engine to programmatically identify matching records and automatically merge; Discard seemingly similar but different records from being consolidated.

Responsibilities: -

- Architected a comprehensive MDM solution to consolidate and unify data from multiple systems into a centralized data lake, ensuring a single, accurate view of organizational data.
- Integrated various applications across the organization, facilitating seamless data sharing, improving data quality, and enabling better decision-making.
- Designed and developed a Match Engine to programmatically identify matching records across different data sources, improving the accuracy and efficiency of data consolidation.
- Automated the merging of duplicate records and implemented logic to discard similar but distinct records, preventing incorrect data aggregation and ensuring data integrity.
- Enhanced data accuracy by ensuring that only verified, relevant data was merged, while eliminating redundancy and inconsistencies across systems.
- Implemented data governance policies to maintain high-quality standards and prevent errors during data integration.
- Ensured the solution could scale to accommodate growing datasets while maintaining performance and data consistency.
- Facilitated smooth integration with legacy systems and third-party applications, ensuring a seamless flow of data into the centralized data lake

Environment: Java 8, Springboot , MySQL, Junit, Jmeter, Swagger, Postman, Github, Jenkins, Apache Lucene

Project:

Built a portal for The Jockey Club to maintain various workflows of this commercial horse racing organization.

- Led the design and development of a comprehensive portal to manage and automate various workflows within the organization, including horse racing events, registrations, and commercial operations.
- Focused on creating an intuitive user interface for seamless interaction, ensuring that users could easily manage complex processes.
- Called the Restful web services calls for POST, PUT, DELETE and GET methods.
- Have worked on AWS to fetching the pictures files from AWS to UI.
- Used React flux to polish the data and for single directional flow.
- Used Object Oriented Programming concepts to develop UI components that could be reused across the Web Application.
- Automated key business processes such as event scheduling, horse registrations, and race management, reducing manual efforts and increasing operational efficiency.
- Enabled real-time updates and notifications, ensuring timely communication and reducing delays in workflow execution.
- Integrated the portal with existing systems, databases, and third-party applications used by The Jockey Club, ensuring smooth data flow and consistency across platforms.
- Provided a customizable solution that allowed The Jockey Club to adjust workflows and processes based on evolving needs and requirements.
- Focused on delivering a user-friendly experience, ensuring the portal was accessible to different user roles and provided appropriate functionality for administrators, event managers, and participants.

- Incorporated mobile-friendly design principles, enabling access to the portal on various devices for greater flexibility and convenience.

Environment: Java 8, Springboot , MySQL, Junit, Jmeter, Swagger, Postman, Github, Jenkins

Siemens Corporate Research – Bangalore ,India

Jan 2011 to Jul 2015

Senior full stack developer

Project:

- Developed, maintained and supported single handedly COSMOS, code quality monitoring dashboard - Empirical research in Program Analysis and Code Quality
- Solely developed, maintained, and supported COSMOS, a comprehensive code quality monitoring dashboard, to track and improve code health across multiple projects.
Integrated various program analysis tools to provide real-time insights into code quality, ensuring high standards of maintainability and performance.
- Developed Algorithms for Incremental Static Analysis: Code duplication, Code refactoring and coding guidelines' violations
- Designed and implemented algorithms for incremental static analysis, detecting issues such as code duplication, code refactoring opportunities, and coding guideline violations.
- Improved overall code quality by automating the detection of common issues and providing actionable insights for developers to refactor code efficiently.
- Trainings, Reviews and Audits for Non-functional requirements, code quality, best practices, compliance, tools & processes